

EduQuiz AI – AI-Powered Syllabus Quiz Generator using IBM watsonx.ai and LangFlow

1. Abstract

EduQuiz AI is an AI-powered educational assistant developed using IBM watsonx.ai, IBM Granite foundation models, and LangFlow. The system automatically generates learning modules, multiple-choice questions (MCQs), correct answers, and explanations from any user-provided educational topic. The project aims to reduce the time required for creating assessments and improve learning through AI-generated educational content.

2. Problem Statement

Creating educational quizzes and learning materials manually is time-consuming for both students and educators. There is a need for an intelligent system that can automatically generate structured learning content and assessments from a given topic.

3. Objectives

- Generate learning modules from any educational topic.
- Create multiple-choice questions automatically.
- Provide correct answers and explanations.
- Support multiple academic subjects.
- Improve learning efficiency through AI-generated content.

4. Proposed Solution

EduQuiz AI uses IBM Granite foundation models hosted on IBM watsonx.ai and orchestrated through LangFlow. Users provide a topic such as Machine Learning, Python Programming, or Database Management Systems. The system processes the topic and generates structured learning modules, MCQs, correct answers, and explanations.

5. Technologies Used

- IBM watsonx.ai
- IBM Granite Foundation Models
- LangFlow
- Prompt Engineering
- Generative AI

- Python

6. System Architecture

User Topic

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LangFlow Chat Input

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Prompt Template

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IBM Granite Foundation Model

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Quiz Generation Engine

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Learning Modules + MCQs + Answers + Explanations

7. LangFlow Components Used

1. Chat Input – Receives user topic.
2. Prompt Template – Formats instructions for quiz generation.
3. IBM Watsonx AI Model – Generates educational content using Granite models.
4. Chat Output – Displays generated quizzes and explanations.

8. Role of Agentic AI

Agentic AI enables the system to understand user topics, generate structured learning content, create quizzes, and provide explanations automatically. IBM Granite models perform reasoning and content generation, making the solution intelligent and adaptive.

9. Sample Output

Input Topic:

Machine Learning

Output:

- 5 Learning Modules
- 10 Multiple Choice Questions
- Correct Answers
- Detailed Explanations

10. Novelty and Uniqueness

- AI-powered quiz generation

- Automated learning module creation
- Instant answer explanations
- Multi-subject support
- IBM Granite integration
- Easy-to-use LangFlow workflow

11. Future Scope

- Difficulty levels (Easy, Medium, Hard)
- PDF export functionality
- Upload notes/PDFs and generate quizzes
- Multi-language support
- Voice-based quiz generation
- Student performance analytics

12. Conclusion

EduQuiz AI demonstrates how IBM watsonx.ai and Granite foundation models can be used to automate educational content generation. The project provides an efficient, scalable, and intelligent solution for creating quizzes and learning resources, helping both students and educators save time and improve learning outcomes.

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